

Day 20 Student Handout

Geometry Summer Credit | Area of Polygons, Composite Figures, and Coordinate Area

Name:	Date:	Period:
-------	-------	---------

Learning Targets

I can find the area of polygons and composite figures. I can use coordinate grids to find perimeter and area. I can choose an efficient strategy before calculating.

Lesson 20A: Area Formulas and Composite Figures

Warm-Up

1.	Find the area of a rectangle with length 14 and width 9.
2.	Find the area of a triangle with base 18 and height 7.
3.	Find the area of a trapezoid with bases 10 and 16 and height 6.

Core Idea

Composite area means a shape is built from simpler shapes. Add areas for pieces that are included. Subtract areas for holes or missing pieces.

Problem	Work / notes
A composite figure is a 12 by 8 rectangle with a 4 by 3 rectangle removed. Find the area.	Large area: Removed area: Composite area:
A figure is a rectangle 10 by 6 with a right triangle attached. The triangle has legs 6 and 8. Find total area.	Rectangle area: Triangle area: Total area:
A regular-looking patio is a trapezoid with bases 18 ft and 26 ft and height 12 ft. Find the area.	Formula: Substitute: Area:

Day 20 Student Handout

Lesson 20B | Coordinate Area and Perimeter

Area on the Coordinate Plane

Strategy First

On graph paper, use horizontal and vertical distances to create rectangles, triangles, and trapezoids. For perimeter, use side lengths; if a side is diagonal, use the distance formula.

Problem	Work / notes
Rectangle A(1, 2), B(9, 2), C(9, 7), D(1, 7). Find area and perimeter.	Length: Width: Area: Perimeter:
Triangle P(0, 0), Q(8, 0), R(8, 6). Find area and perimeter.	Base: Height: Hypotenuse: Area: Perimeter:
A coordinate composite figure fits inside a 10 by 8 rectangle with two right triangles removed, each with legs 3 and 4. Find area.	Bounding rectangle: Removed triangles: Remaining area:

4.	Explain how coordinate geometry connects today's lesson to slope, distance, and Algebra 2 readiness.
----	--

5.	What is one mistake students make when finding composite area, and how can you avoid it?
----	--